sEMG Muscle Activation Report

Participant P1 · Gesture_1 · April 17, 2026

Overall Activation	High
Temporal Profile	Ramp-down
Consistency	Consistent
Dominant Sensors	S4
Weakest Sensors	S7
Assessment Confidence	0.95

Clinical Summary: Sustained contraction shows high flexor activation, particularly in S4. The ramp-down profile in S4 suggests potential fatigue or loss of force maintenance over the 20-minute session. Co-contraction ratio is below the 0.5 threshold, indicating minimal antagonistic interference.

Key Findings:
- Ramp-down profile in S4 and S7
- High flexor-extensor imbalance

Session Overview

Session Duration	1207.6 s (20.1 min)
Sensors	8 (S1–S4 flexor, S5–S8 extensor)
Sampling Rate	1259.3 Hz
Session Mean RMS	0.008686 mV
Feature Source	raw_signal

Per-Sensor Activation Analysis

Muscle activity was recorded across 8 sensors. Sensors S1–S4 are positioned on the palmar/flexor compartment and S5–S8 on the dorsal/extensor compartment. RMS (Root Mean Square) reflects overall activation intensity; MNF and MDF (Mean and Median Frequency) characterise the spectral content of the EMG signal; WAMP (Willison Amplitude) indicates motor-unit firing activity.

Sensor	Mean RMS (mV)	CV	MNF (Hz)	MDF (Hz)	WAMP	Level	Profile
S1	0.004786	0.170	105.6	80.2	22	Moderate	Sustained
S2	0.006875	0.067	106.5	85.0	126	High	Sustained
S3	0.015222	0.071	98.1	74.8	2128	Very High	Sustained
S4	0.023874	0.134	91.1	79.9	4176	Very High	Ramp-down
S5	0.005268	0.129	94.8	65.7	14	Moderate	Sustained
S6	0.005849	0.149	87.2	54.6	107	Moderate	Sustained
S7	0.003284	0.144	120.5	92.5	12	Moderate	Ramp-down
S8	0.004332	0.219	120.7	94.5	66	Moderate	Sustained

Flexor sensors (S1–S4) shown with blue labels; extensor sensors (S5–S8) with pink.

Notable Findings per Sensor

S4: Primary driver, significant amplitude decay

S8: Borderline consistency

Spatial Activation Patterns

Cross-channel metrics characterise how muscle activation is distributed across the forearm during the recorded gesture. A **high concentration index** indicates focal activation (one or two dominant sensors); a **low value** suggests diffuse activation. The **flexor/extensor balance** reveals which compartment drives the gesture. **Pattern stability** near 1.0 indicates the spatial pattern is maintained throughout the session, with no substantial load redistribution.

Metric	Value	Interpretation
Dominant Sensor	S4	Primary contributor
Weakest Sensor	S7	Lowest activity
Dominant-to-Weakest Ratio	7.27×	Activation disparity
Concentration Index	0.201	Focal (near 1) vs Diffuse (near 0.125)
Pattern Stability	0.998	Spatial consistency (1 = stable)
Inter-Channel Correlation	0.379	Crosstalk check (>0.8 = concern)
Co-Contraction Ratio	0.369	Agonist/antagonist balance
Flexor/Extensor Balance	2.73	>1 = flexor-dominant

Overall Temporal Profile	Ramp-down	Session-wide activation trend

Co-contraction: Not detected — Reciprocal activation pattern typical of efficient single-direction movement.

Spatial Compensation: Not detected — Stable spatial activation pattern throughout the session.

Gesture-Pattern Interpretation: Strong flexor dominance (S1-S4) compared to extensor group (S5-S8) with a flexor-extensor balance ratio of 2.73.

Motor-Unit Recruitment

Motor-unit recruitment is inferred from the Willison Amplitude (WAMP) and zero-crossing rate. Elevated WAMP indicates frequent changes in signal amplitude, consistent with the firing of additional (often higher-threshold) motor units as effort or fatigue increases.

Recruitment Detected	Yes
Sensors Affected	S4, S3

Evidence: High WAMP values in S4 (4176.3) and S3 (2127.5) indicate significant motor unit recruitment in the flexor group.

Visualisation

The figure below consolidates the session analysis: per-sensor RMS amplitudes, spatial activation map, RMS time series for dominant sensors, co-contraction ratio, and frequency-content (MNF/MDF) trends over the 20-minute session.

Methodology & Limitations

Signal Processing

Raw surface EMG signals from 8 Delsys Avanti sensors were band-pass filtered (20–450 Hz, Butterworth 4th order) to attenuate motion artefacts and high-frequency noise. Twelve features were extracted per sensor in consecutive 20-second windows: RMS, MAV, IEMG, Waveform Length, Zero Crossings, Slope Sign Changes, Willison Amplitude, Peak Amplitude, Variance, Mean Frequency, Median Frequency, and Total Power. Cross-channel metrics (activation map, concentration index, pattern stability, co-contraction ratio, flexor/extensor balance) were computed from the per-sensor RMS time series.

Clinical Interpretation

Feature summaries were passed to a large language model (gemini-3.1-flash-lite-preview) along with a structured set of clinical rules (activation level thresholds, gesture patterns, co-contraction criteria, motor-unit recruitment indicators) and retrieval-augmented research context drawn from a knowledge base of sEMG literature (Konrad 2005, Phinyomark 2012, Vigotsky 2018, Nazmi 2016, Yang 2024 EMGBench).

Limitations

• **No %MVC normalisation:** Amplitudes are absolute (mV); direct cross-subject comparisons of amplitude are unreliable without a reference contraction.
• **Sustained-contraction data:** Because the gesture is held continuously, onset and duty-cycle metrics are not reported.
• **No ground-truth validation:** Findings are descriptive. Clinical decisions should consider additional context (force measurement, movement observation, patient history).
• **LLM output:** Provided as a summarisation aid, not a diagnostic tool. Human-in-the-loop review is required.

Report generated automatically by the sEMG Interpretation Pipeline. For clinical use, please corroborate findings with additional assessments.